TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	89.2 ± 0.87
Correlation Stability	88.0 ± 1.26
Distribution Consistency	93.0 ± 0.77
Entropy Preservation	92.2 ± 0.40
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.2 ± 1.17
Feature Accuracy	92.2 ± 1.40
Novelty Score	0.9 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	51.881	+0.398	0.8%	OK
bp	76.469	76.906	+0.437	0.6%	OK
sg	1.017	1.017	-0.001	0.1%	OK
al	1.017	1.044	+0.027	2.6%	OK
su	0.450	0.412	-0.038	8.4%	OK
rbc	0.190	0.125	-0.065	34.0%	Drift
pc	0.227	0.216	-0.011	5.0%	OK
pcc	0.106	0.134	+0.028	26.7%	Drift
ba	0.056	0.037	-0.018	32.5%	Drift
bgr	148.037	153.244	+5.207	3.5%	OK
bu	57.426	61.299	+3.873	6.7%	OK
sc	3.072	3.166	+0.093	3.0%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	89.2 ± 0.87
Correlation Stability (*2)	88.0 ± 1.26
Distribution Consistency (*3)	93.0 ± 0.77
Entropy Preservation (*4)	92.2 ± 0.40
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.2 ± 1.17
Feature Accuracy (*10)	92.2 ± 1.40
Novelty Score (*11)	0.9 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	0.88375	0.89625	-0.013749999999999984	
F1	0.92384446063128	0.9058381081393019	0.9136809910473026	-0.01016346958397748	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	0.925	0.985	0.006249999999999978	
F1	0.9805471229310466	0.9377058302637316	0.9874678877537523	0.006920764822705749	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	0.9275	0.9725000000000001	-0.009999999999999976	
F1	0.9848619694663338	0.9386927223043209	0.9754365559669417	-0.009425413499392143	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.74875	0.7150000000000001	0.6174999999999999	-0.13125	
F1	0.7292474649295361	0.7903288252140375	0.7154912801572975	-0.0137561847722386	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	0.8712500000000001	0.9075	-0.045000000000000005	*
F1	0.9587663012688215	0.8770150660744106	0.9155818488456674	-0.043184452423154183	*

Model: GB

TabuLLM: AI Benchmark Report

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9887500000000001	0.9337500000000001	0.9825000000000002	-0.006249999999999989	
F1	0.990128320402674	0.940236762781234	0.9854395411413599	-0.004688779261314358	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9462499999999998	0.85625	0.9475	0.0012500000000000067	
F1	0.9525102387557931	0.872913857227109	0.9550137209621434	0.002503482206350493	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.99875	0.9662500000000002	0.99	-0.008750000000000003	
F1	0.9989010989010989	0.9718033498215461	0.9918478081058726	-0.007053290795226286	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	0.6024999999999999	0.6024999999999999	0	
F1	0.7507214882964476	0.7507214882964476	0.7507214882964476	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	0.635	0.6825	0.07	
F1	0.5885198185573248	0.7456479651954989	0.7309479029490322	0.14242808439170748	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	61.0	80.0	1.02	2.0	0.0	0.0	0.0	0.0	0.0	97.0
2	67.0	60.0	1.025	2.0	0.0	0.0	0.0	1.0	0.0	105.0
3	51.0	70.0	1.01	0.0	0.0	0.0	1.0	0.0	0.0	171.0
4	70.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	125.0
5	44.0	90.0	1.0173131672597866	1.0	0.0	0.0	0.0	0.0	0.0	140.0
6	35.0	80.0	1.015	4.0	0.0	0.0	1.0	0.0	0.0	139.0
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	38.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	121.0
2	23.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	113.0
3	69.0	60.0	1.02	0.0	0.0	0.0	1.0	0.0	0.0	235.0
4	59.0	60.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	121.0
5	20.0	60.0	1.025	2.0	0.0	0.0	0.0	0.0	0.0	232.0
6	5.0	90.0	1.0171908127208478	2.0	2.0	0.0	0.0	0.0	0.0	207.0
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	62.0	70.0	1.015	2.0	0.0	0.0	0.0	0.0	0.0	120.0
2	23.0	70.0	1.02	4.0	0.0	0.0	1.0	1.0	0.0	232.0
3	76.0	90.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	102.0
4	46.0	80.0	1.01	1.0	0.0	0.0	1.0	0.0	0.0	144.0
5	61.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	81.0
6	5.0	80.0	1.015	4.0	0.0	1.0	0.0	0.0	0.0	97.0
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

TabuLLM: AI Benchmark Report

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	76.0	70.0	1.015	0.0	1.0	0.0	0.0	0.0	0.0	132.0
2	55.0	90.0	1.015	4.0	4.0	1.0	1.0	0.0	0.0	424.0
3	42.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	123.0
4	39.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	118.0
5	41.0	80.0	1.0174290780141844	0.0	0.0	0.0	0.0	0.0	0.0	114.0
6	44.0	90.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	123.0
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	56.0	60.0	1.01	0.0	0.0	0.0	0.0	0.0	0.0	95.0
2	12.0	70.0	1.0174014336917563	0.0	0.0	0.0	0.0	0.0	0.0	91.0
3	55.0	70.0	1.01	3.0	2.0	1.0	0.0	0.0	0.0	412.0
4	65.0	90.0	1.025	0.0	0.0	1.0	0.0	0.0	0.0	109.0
5	48.0	80.0	1.0174014336917563	0.0	0.0	0.0	0.0	0.0	0.0	124.0
6	56.0	90.0	1.02	2.0	0.0	0.0	1.0	0.0	0.0	110.0
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

TabuLLM: AI Benchmark Report

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	63.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	246.0
2	45.0	90.0	1.01	2.0	2.0	1.0	1.0	0.0	0.0	122.0
3	79.0	90.0	1.02	3.0	4.0	0.0	0.0	0.0	0.0	371.0
4	29.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	100.0
5	73.0	90.0	1.01	0.0	0.0	0.0	0.0	0.0	1.0	100.0
6	73.0	80.0	1.015	4.0	2.0	0.0	1.0	1.0	0.0	165.0
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	61.0	80.0	1.02	1.0	0.0	0.0	0.0	0.0	0.0	107.0
2	47.0	80.0	1.0187624658460646	0.0	0.0	0.0	0.0	0.0	0.0	209.0
3	60.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	204.0
4	52.0	80.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	463.0
5	70.0	80.0	1.0160153446491547	0.0	3.0	0.0	0.0	0.0	0.0	172.0
6	70.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	92.0
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	39.0	80.0	1.025	2.0	0.0	0.0	0.0	0.0	0.0	137.0
2	71.0	80.0	1.01	0.0	0.0	1.0	0.0	1.0	0.0	237.0
3	60.0	100.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	315.0
4	33.0	70.0	1.01	2.0	0.0	0.0	0.0	1.0	0.0	169.0
5	60.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
6	54.0	90.0	1.02	4.0	0.0	0.0	0.0	0.0	0.0	104.0
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	34.0	90.0	1.015	1.0	0.0	0.0	0.0	0.0	0.0	91.0
2	38.0	90.0	1.01722807017543 86	0.0	0.0	0.0	0.0	0.0	0.0	100.0
3	39.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	88.0
4	60.0	90.0	1.01	4.0	2.0	0.0	1.0	1.0	0.0	424.0
5	36.0	60.0	1.015	0.0	0.0	0.0	0.0	0.0	0.0	118.0
6	48.0	60.0	1.02	0.0	0.0	0.0	1.0	0.0	0.0	248.0
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	58.0	80.0	1.005	2.0	0.0	0.0	0.0	0.0	0.0	360.0
2	45.0	90.0	1.02	1.0	4.0	0.0	0.0	1.0	0.0	133.0
3	39.0	60.0	1.01717081850533 8	3.0	0.0	0.0	0.0	0.0	0.0	93.0
4	50.0	70.0	1.01717081850533 8	0.0	0.0	0.0	0.0	0.0	0.0	107.0
5	35.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	138.0
6	34.0	80.0	1.015	4.0	4.0	0.0	0.0	1.0	0.0	137.0
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0